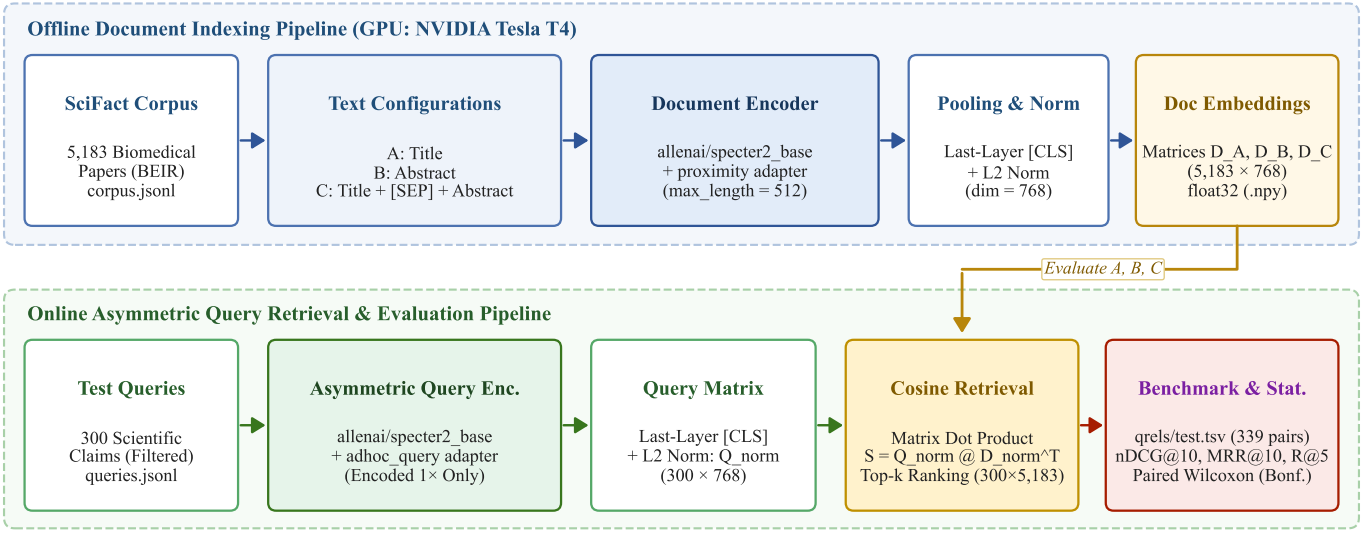

Offline Document Indexing Pipeline (GPU: NVIDIA Tesla T4)
SciFact Corpus
5,183 Biomedical Papers (BEIR) corpus.jsonl
Text Configurations
A: Title
B: Abstract
C: Title + [SEP] + Abstract
Document Encoder
allenai/specter2_base + proximity adapter (max_length = 512)
Pooling & Norm
Last-Layer [CLS] + L2 Norm (dim = 768)
Doc Embeddings
Matrices D_A, D_B, D_C (5,183 × 768) float32 (.npy)
Evaluate A, B, C
Online Asymmetric Query Retrieval & Evaluation Pipeline
Test Queries
300 Scientific Claims (Filtered) queries.jsonl
Asymmetric Query Enc.
allenai/specter2_base + adhoc_query adapter (Encoded 1× Only)
Query Matrix
Last-Layer [CLS] + L2 Norm: Q_norm (300 × 768)
Cosine Retrieval
Matrix Dot Product S = Q_norm @ D_norm^T Top-k Ranking (300×5,183)
Benchmark & Stat.
qrels/test.tsv (339 pairs) nDCG@10, MRR@10, R@5 Paired Wilcoxon (Bonf.)